

Your grade: 100%

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Next item →

1. Which one of the following is an injective function?

1 / 1 point

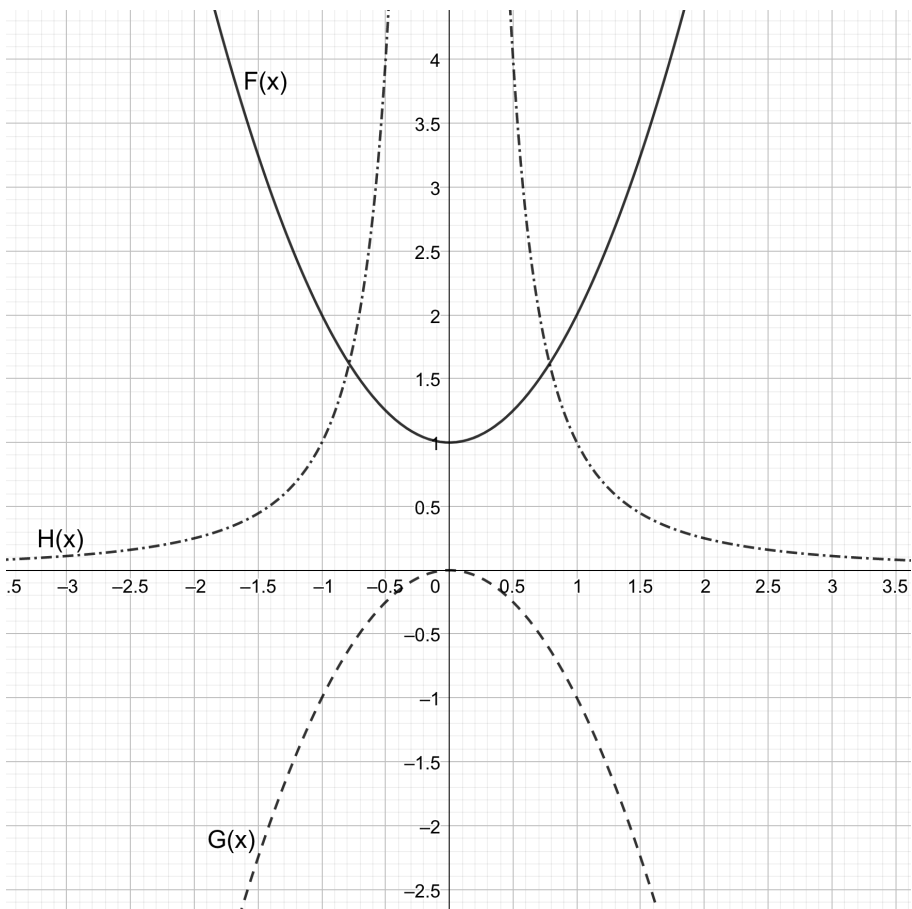
- ☐ $f: \mathbb{Z} \rightarrow \mathbb{Z}^+$ with $f(x) = |x|$
- ☒ $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}$ with $f(x) = |x|$
- ☐ $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = |x|$
- ☐ $f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(x) = |x|$

Nice work

for all $a, b \in \mathbb{Z}^+$ if $a \neq b$ then $|a| = a \neq b = |b|$.

2. The following graph shows three curves labelled F(x), G(x) and H(x). Which label represents the curve of the function $f(x) = x^2 + 1$?

1 / 1 point



- ☐ None of these answers
- ☒ F(x)
- ☐ G(x)
- ☐ H(x)

Nice work

$f(x)$ is a quadratic function which has no real roots and cuts the y-axis at $(0, 1)$, when $x = 0, y = 1$.

3. Which one of the following functions is surjective?

1 / 1 point

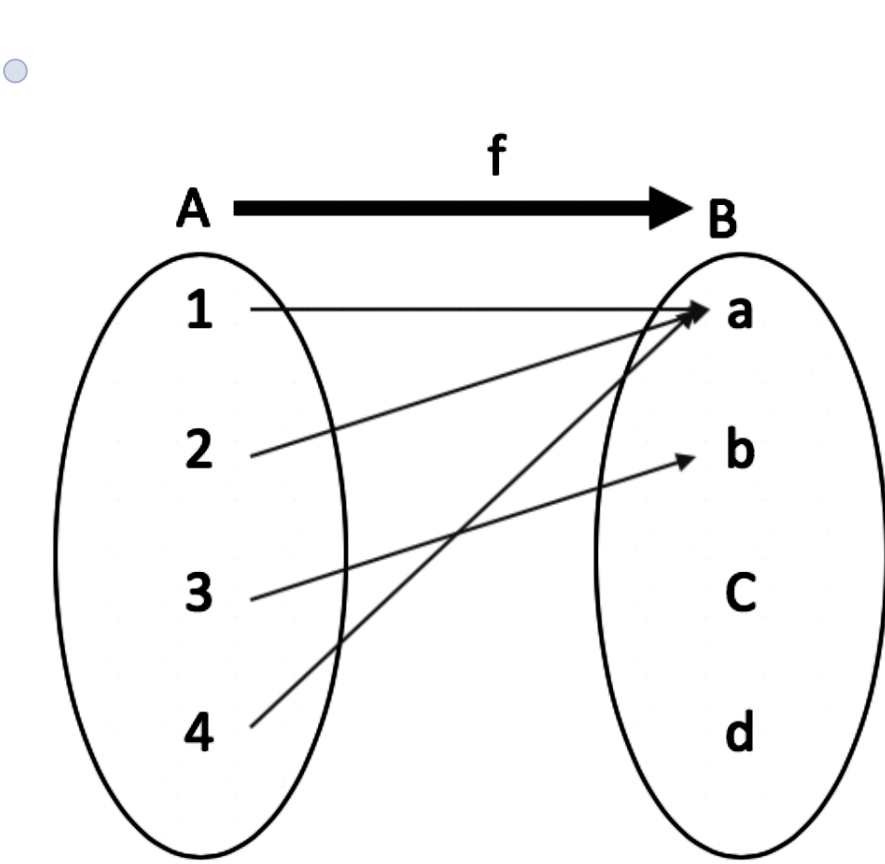
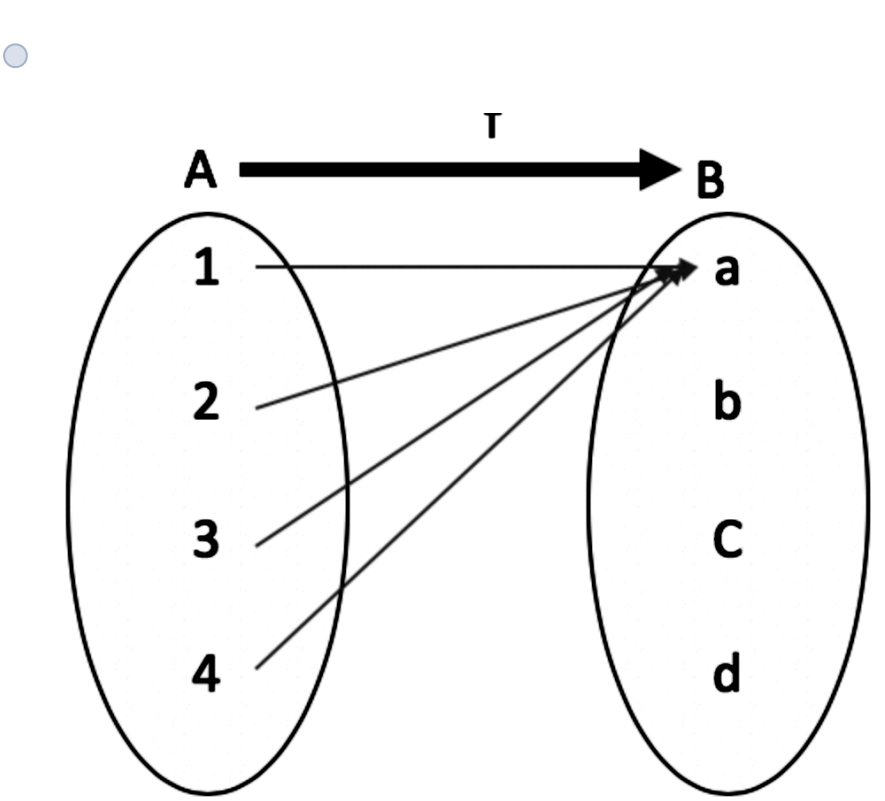
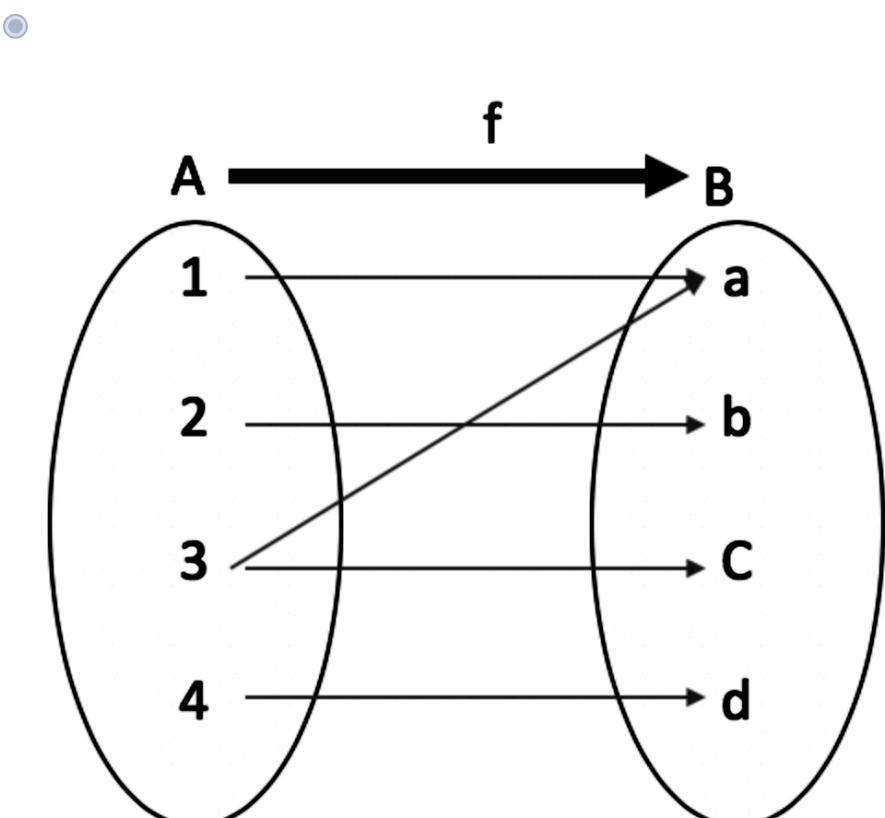
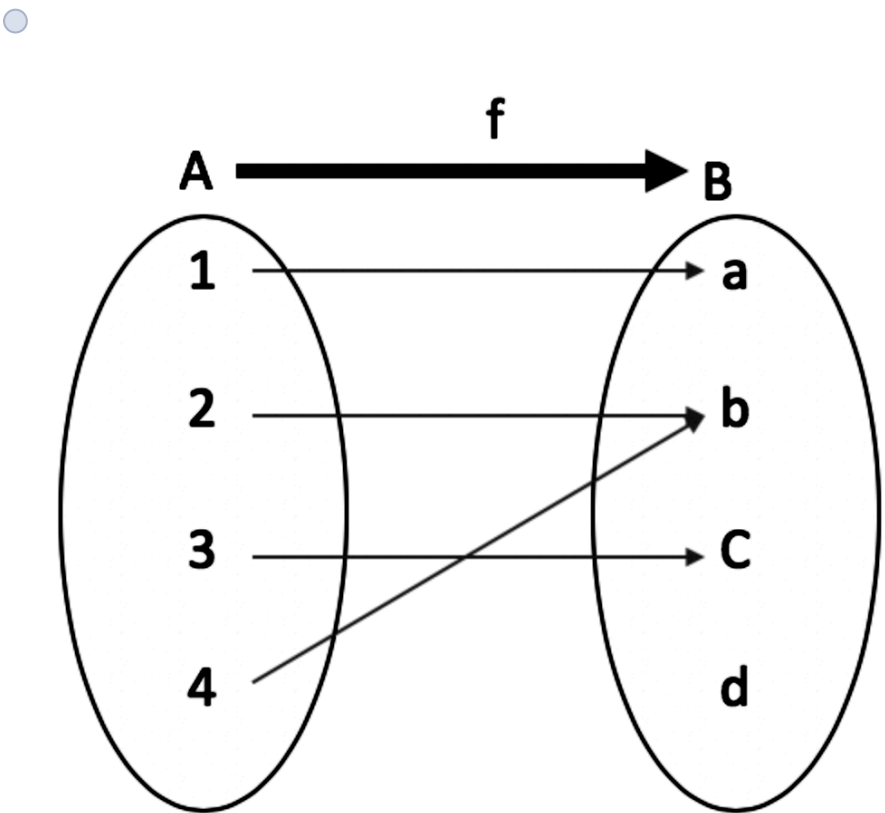
- ☐ $f: \mathbb{N} \rightarrow \mathbb{N}$ with $f(x) = 3x + 2$
- ☐ $f: \mathbb{Z} \rightarrow \mathbb{Z}$ with $f(x) = 3x + 2$
- ☒ $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = 3x + 2$
- ☐ $f: \mathbb{Z}^+ \rightarrow \mathbb{R}$ with $f(x) = 3x + 2$

Nice work

$f(x) = y \implies 3x + 2 = y \implies x = \frac{y-2}{3}$, hence for $y \in \mathbb{R}$ there exists $x = \frac{y-2}{3} \in \mathbb{R}$ such that $f(x) = y$.

4. Which one of the following is not a function?

1 / 1 point



Nice work

f is not a function. $3 \in A$ has more than one image in B .

5. Let $f: \mathbb{R}^+ \rightarrow \mathbb{R}$ with $f(x) = \ln(3x)$. Which one of the following functions is equal to $f^{-1}(x)$?

1 / 1 point

- ☐ $\frac{x}{3}$
- ☒ $\frac{e^x}{3}$
- ☐ $\frac{e^x}{3}$
- ☐ $\frac{x}{3}$

Nice work

$y = \ln(3x) \implies 3x = e^y \implies x = \frac{e^y}{3}$